

## CAIRNS, Australia

WMO: 942870

Lat: 16.8736S

Lon: 145.7450E

Elev: 8

StdP: 101.23

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 13.3       | 14.9 | 6.3                           | 5.9 | 20.9 | 8.3 | 6.8 | 19.9 | 10.9                  | 24.2 | 10.2 | 24.1 | 2.1                   | 280  | 0.423 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 6.8                    | 32.9            | 25.0 | 32.0 | 24.8 | 31.2 | 24.7 | 27.0                | 30.4 | 26.5 | 29.9 | 26.1 | 29.5 | 5.6                  | 120  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 26.1                            | 21.5 | 29.0 | 25.6 | 20.8 | 28.5 | 25.1 | 20.3 | 28.1 | 85.2          | 30.8 | 83.1 | 30.1 | 81.3 | 29.6 | 29.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |      |     |
| (4) 10.1          | 9.2  | 8.4 | DB                         | 10.9 | 35.8               | 1.5 | 1.7                                                | 9.8 | 37.1       | 8.9 | 38.1       | 8.0 | 39.1       | 6.9 | 40.3 | (4) |
| (5)               |      |     | WB                         | 9.5  | 28.2               | 1.1 | 0.5                                                | 8.7 | 28.6       | 8.1 | 28.9       | 7.5 | 29.2       | 6.7 | 29.6 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 25.0  | 27.6  | 27.5  | 26.9  | 25.6  | 23.9  | 22.3  | 21.5  | 21.9  | 23.5  | 25.2  | 26.4  | 27.4 | (6)  |
| (7)  |                                                                                  | DBStd       | 2.62  | 1.22  | 1.29  | 1.14  | 1.28  | 1.47  | 1.79  | 1.84  | 1.65  | 1.52  | 1.34  | 1.15  | 1.32 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD18.3     | 3     | 0     | 0     | 0     | 0     | 0     | 1     | 2     | 1     | 0     | 0     | 0     | 0    | (9)  |
| (10) |                                                                                  | CDD10.0     | 5458  | 544   | 491   | 522   | 469   | 432   | 369   | 356   | 369   | 404   | 469   | 493   | 540  | (10) |
| (11) |                                                                                  | CDD18.3     | 2420  | 286   | 257   | 264   | 219   | 173   | 121   | 99    | 112   | 154   | 211   | 243   | 282  | (11) |
| (12) |                                                                                  | CDH23.3     | 19143 | 2948  | 2592  | 2388  | 1560  | 857   | 408   | 317   | 477   | 928   | 1566  | 2186  | 2917 | (12) |
| (13) | CDH26.7                                                                          | 5399        | 1022  | 871   | 699   | 368   | 116   | 24    | 12    | 37    | 155   | 395   | 666   | 1036  | (13) |      |
| (14) | Wind                                                                             | WSAvg       | 4.4   | 3.7   | 3.6   | 4.4   | 5.0   | 4.9   | 4.9   | 4.6   | 4.3   | 4.3   | 4.1   | 3.8   | (14) |      |
| (15) | Precipitation                                                                    | PrecAvg     | 2012  | 390   | 454   | 407   | 202   | 90    | 45    | 30    | 22    | 31    | 45    | 94    | 181  | (15) |
| (16) |                                                                                  | PrecMax     | 3171  | 1417  | 1307  | 1113  | 635   | 322   | 180   | 145   | 81    | 101   | 204   | 372   | 919  | (16) |
| (17) |                                                                                  | PrecMin     | 718   | 86    | 30    | 28    | 12    | 3     | 3     | 3     | 0     | 0     | 0     | 3     | 27   | (17) |
| (18) |                                                                                  | PrecStd     | 540   | 266   | 271   | 267   | 133   | 66    | 39    | 25    | 20    | 32    | 45    | 86    | 168  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 34.2  | 34.5  | 32.8  | 31.2  | 29.7  | 28.2  | 27.9  | 28.8  | 30.0  | 31.2  | 32.2  | 33.8 | (19) |
| (20) |                                                                                  |             | MCWB  | 25.3  | 25.8  | 25.3  | 23.7  | 22.9  | 21.9  | 21.0  | 20.5  | 20.6  | 22.7  | 23.8  | 25.0 | (20) |
| (21) |                                                                                  | 2%          | DB    | 32.4  | 32.6  | 31.6  | 30.2  | 28.8  | 27.2  | 26.9  | 27.7  | 29.1  | 30.2  | 31.2  | 32.3 | (21) |
| (22) |                                                                                  |             | MCWB  | 25.2  | 25.5  | 24.9  | 23.4  | 22.3  | 21.2  | 20.3  | 20.2  | 20.9  | 22.6  | 23.4  | 24.6 | (22) |
| (23) |                                                                                  | 5%          | DB    | 31.6  | 31.7  | 30.9  | 29.7  | 28.0  | 26.6  | 26.0  | 26.9  | 28.2  | 29.5  | 30.5  | 31.7 | (23) |
| (24) |                                                                                  |             | MCWB  | 25.1  | 25.4  | 24.8  | 23.2  | 21.8  | 20.7  | 19.5  | 19.7  | 20.6  | 22.4  | 23.3  | 24.6 | (24) |
| (25) |                                                                                  | 10%         | DB    | 30.9  | 30.9  | 30.1  | 29.0  | 27.1  | 25.8  | 25.1  | 26.0  | 27.6  | 28.9  | 29.9  | 30.9 | (25) |
| (26) |                                                                                  |             | MCWB  | 25.1  | 25.4  | 24.5  | 23.0  | 21.4  | 20.3  | 19.1  | 19.1  | 20.5  | 22.1  | 23.2  | 24.6 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 27.4  | 27.8  | 27.2  | 26.1  | 24.6  | 23.7  | 22.7  | 22.9  | 24.2  | 25.3  | 26.4  | 27.2 | (27) |
| (28) |                                                                                  |             | MCDWB | 30.7  | 31.2  | 30.5  | 28.8  | 27.9  | 26.5  | 25.2  | 25.7  | 27.5  | 28.8  | 29.7  | 30.8 | (28) |
| (29) |                                                                                  | 2%          | WB    | 26.8  | 26.9  | 26.4  | 25.3  | 23.7  | 22.6  | 21.7  | 21.8  | 23.2  | 24.6  | 25.6  | 26.6 | (29) |
| (30) |                                                                                  |             | MCDWB | 30.2  | 30.3  | 29.7  | 28.2  | 26.7  | 25.5  | 24.9  | 25.4  | 26.6  | 28.1  | 29.0  | 30.2 | (30) |
| (31) |                                                                                  | 5%          | WB    | 26.3  | 26.4  | 25.9  | 24.7  | 23.1  | 22.0  | 21.1  | 21.0  | 22.4  | 23.9  | 25.0  | 26.1 | (31) |
| (32) |                                                                                  |             | MCDWB | 29.7  | 29.7  | 29.0  | 27.6  | 26.1  | 24.9  | 24.2  | 24.7  | 26.1  | 27.5  | 28.5  | 29.6 | (32) |
| (33) |                                                                                  | 10%         | WB    | 25.9  | 26.0  | 25.4  | 24.1  | 22.5  | 21.3  | 20.5  | 20.4  | 21.7  | 23.3  | 24.4  | 25.5 | (33) |
| (34) |                                                                                  |             | MCDWB | 29.2  | 29.3  | 28.4  | 27.1  | 25.6  | 24.3  | 23.7  | 24.4  | 25.6  | 27.0  | 28.0  | 29.1 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 6.8   | 6.5   | 6.5   | 6.7   | 7.0   | 7.1   | 8.0   | 8.6   | 8.8   | 7.9   | 7.4   | 7.1  | (35) |
| (36) |                                                                                  |             | MCDBR | 7.9   | 7.9   | 7.7   | 7.8   | 7.5   | 7.7   | 8.3   | 9.4   | 9.3   | 8.3   | 8.1   | 8.1  | (36) |
| (37) |                                                                                  |             | MCWBR | 3.2   | 3.3   | 3.1   | 3.2   | 3.3   | 3.5   | 3.6   | 3.9   | 3.7   | 3.1   | 3.0   | 3.0  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 6.8   | 6.7   | 6.3   | 6.0   | 6.2   | 6.2   | 6.7   | 7.7   | 7.8   | 7.2   | 6.8   | 6.9  | (38) |
| (39) |                                                                                  |             | MCWBR | 3.3   | 3.3   | 3.1   | 3.0   | 3.1   | 3.2   | 3.3   | 3.6   | 3.7   | 3.3   | 3.2   | 3.2  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.393 | 0.394 | 0.392 | 0.379 | 0.366 | 0.359 | 0.342 | 0.349 | 0.377 | 0.395 | 0.395 | 0.409 | (40) |      |
| (41) |                                                                                  | taud        | 2.574 | 2.573 | 2.566 | 2.582 | 2.588 | 2.591 | 2.624 | 2.589 | 2.488 | 2.463 | 2.502 | 2.490 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 952   | 941   | 922   | 898   | 875   | 864   | 892   | 914   | 921   | 929   | 943   | 936   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 107   | 106   | 104   | 96    | 89    | 86    | 85    | 94    | 110   | 117   | 115   | 117   | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 5.91  | 5.68  | 5.25  | 4.83  | 4.26  | 3.80  | 4.17  | 5.07  | 6.11  | 6.63  | 6.78  | 6.50  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.49  | 0.65  | 0.46  | 0.47  | 0.37  | 0.35  | 0.31  | 0.39  | 0.32  | 0.43  | 0.54  | 0.72  | (45) |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | N/A   | N/A     | N/A    | +0.31   | N/A   | +0.20 | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (6 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page